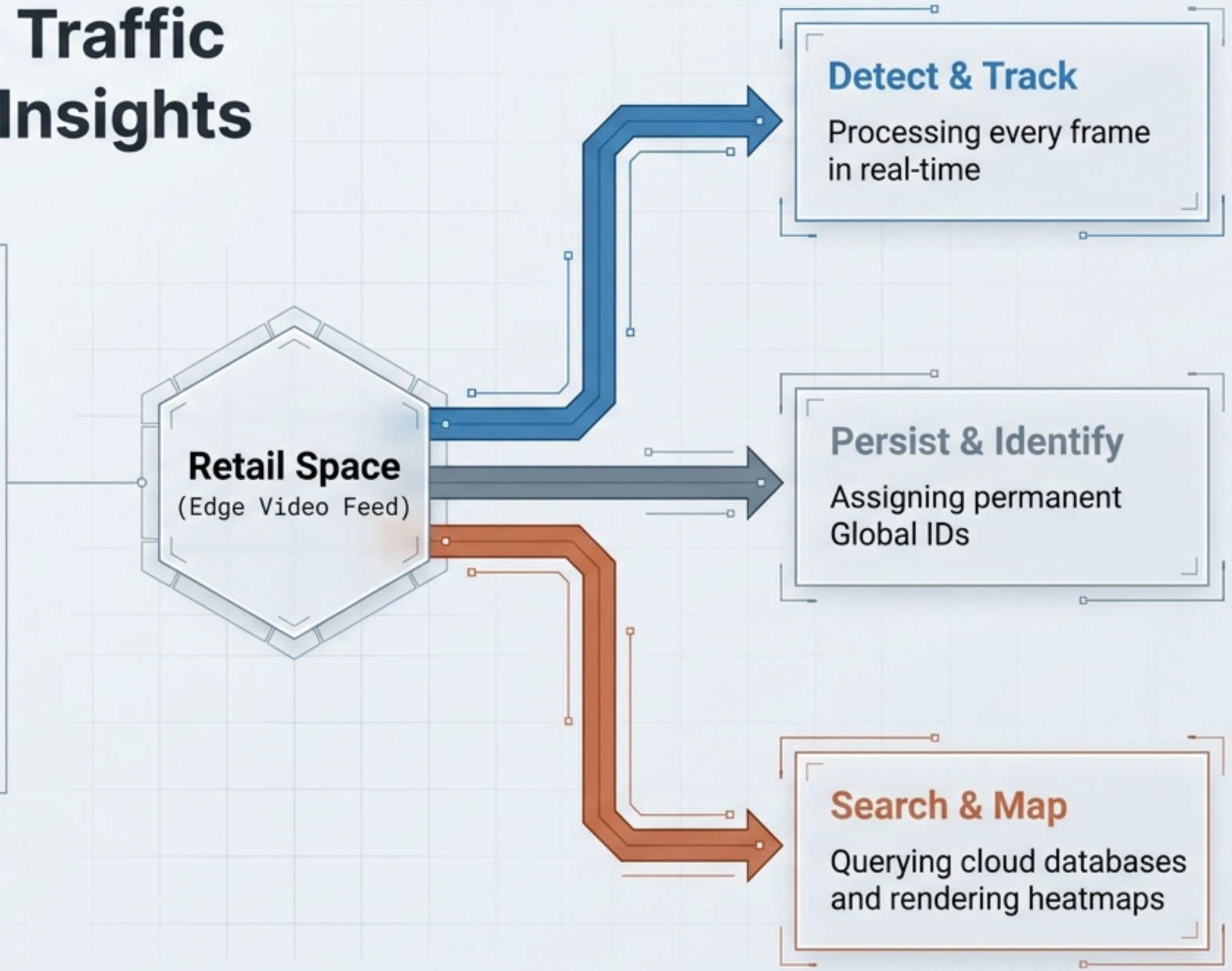
Tracking Retail Foot Traffic from Edge to Cloud Insights

System Environment: Python 3.12, PyTorch

Hardware Target: 2x `NVIDIA Tesla T4 GPUs (16 GB VRAM each)` natively on `Kaggle`

Core Output: Converts raw pixel arrays of moving people into persistent, searchable metadata profiles with spatial movement history.



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Merging Two Modalities to Establish Absolute Ground Truth

Body Re-ID (The Workhorse)

Model:	CLIP-ReID (1280-D Vector)
Strength:	Works at any distance/angle; does not require face visibility.
Weakness:	Fails with similar clothing (e.g., store uniforms).
System Role:	Primary continuous identification.

Face Recognition (The VIP Override)

Model:	ArcFace (512-D Vector)
Strength:	Extremely accurate; highly discriminative identity ground truth.
Weakness:	Requires close-up, well-lit, front-facing captures.
System Role:	Ultimate authority; retroactively corrects body-based assignments.

Key Engineering Insight: Face recognition acts as the ultimate ground truth, capable of retroactively triggering an Identity Merge to rewrite mistaken body-track assignments.

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Five Specialized Neural Networks Powering the Pipeline

cuda:0

RT-DETR-L (Person Detection)

~130MB. Uses a Transformer decoder to eliminate Non-Maximum Suppression (NMS), ensuring predictable latency crucial for real-time tracking. Filters for COCO class 0.

InsightFace Pack (Face Pipeline)

~326MB total. SCRFD handles bounding boxes/landmarks; ArcFace extracts the 512-D highly discriminative identity vector.

cuda:1 & CPU

YOLOv11s-Pose (Skeleton Quality Gate)

Tracks 17 COCO keypoints. Does NOT identify; acts purely as an entrance filter for body crops.

CLIP-ReID (Body Vectors)

1280-D embedding optimized via a 2-stage frozen CLIP backbone process.

BotSort (Multi-Object Tracker)

Operates on CPU/RAM. Uses a Kalman Filter and OSNet appearance tracking to assign temporary frame-by-frame IDs.

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Filtering Raw Pixel Arrays Through Strict Quality Gates



Raw Video Frame
[640x480, 30fps]

Steps 1-2: Detect & Track
RT-DETR finds bodies; BotSort assigns a temporary track_id.
`DET_CONF_THRESH > 0.35`



REJECTED: No person / Lost track

Step 3: Confidence Gate
REID_CONF_THRESHOLD. Drops detections < 0.75 to eliminate ghosts.
`REID_CONF_THRESH = 0.75`



REJECTED: Low confidence (<0.75)

Step 4: Pose Gate
MIN_VISIBLE_RATIO. YOLOv11s-Pose rejects crops where < 50% of the 17 skeleton keypoints are visible.
`MIN_VISIBLE_RATIO = 0.50`



REJECTED: Poor pose visibility (<50%)

Step 5: Embed Body
Surviving crops pass through CLIP-ReID.
`CLIP-ReID: 1280-D Vector Output`

Step 6: Spatial Face Pairing
Checks if an InsightFace bounding box exists within the top 40% of the body bounding box.
`FACE_REGION_THRESH = 0.40`



REJECTED: No valid face pairing within body bbox

Steps 7-8: Vector Lookup & ID Resolution
Queries Pinecone, checks softmax thresholds, and assigns the final identity.
`PINE_CONE_THRESH > 0.85`

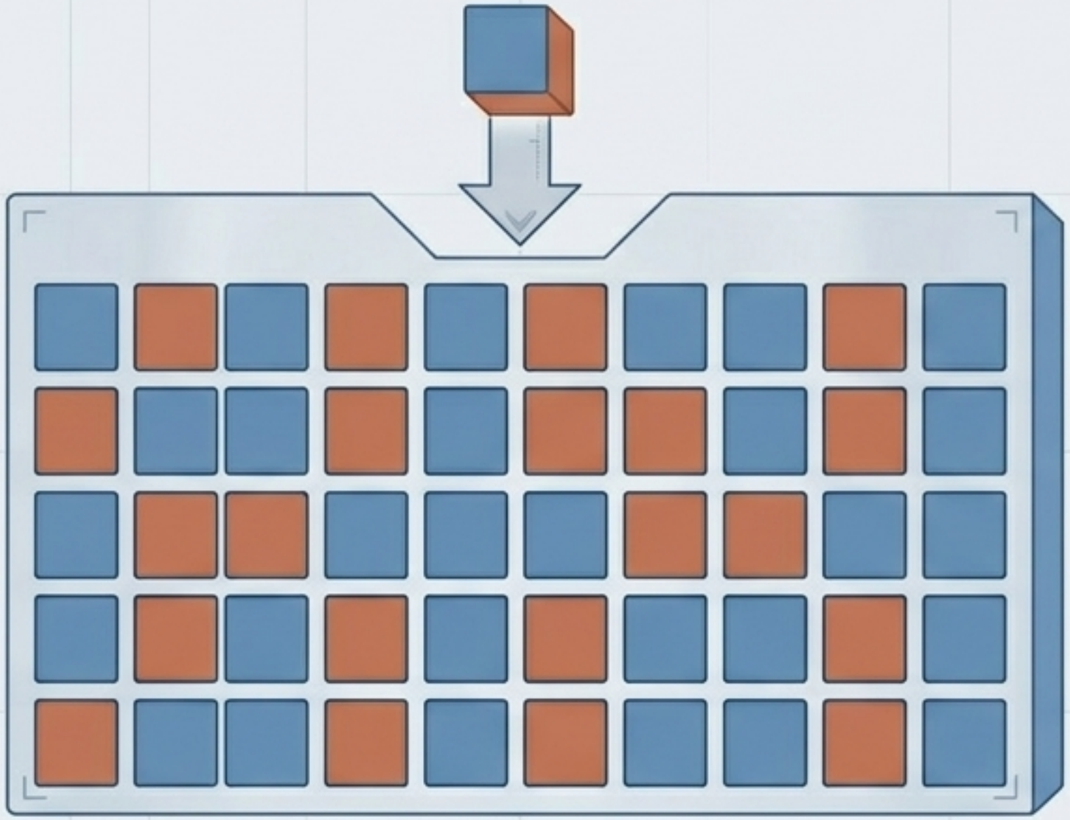


REJECTED: No match / Below resolution threshold

Global ID Sighting
[Resolved Identity: ID-192845]

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Curating a High-Fidelity Memory Within Strict Database Budgets



The Constraint: The Self-Adaptive Gallery allows a maximum of 50 crops per identity (`MAX_IMAGES`).

Drop Cost = Redundancy + Uncertainty

Redundancy:

Evaluated by the maximum `cosine similarity` to any `other crop` of the exact same person. (High score = visually redundant).

Uncertainty:

Evaluated by the `normalized entropy` over `softmax similarities` to all gallery centroids. (High score = ambiguous crop that looks like multiple people).

System Action: The single crop with the highest combined Drop Cost is deleted. It is both uselessly similar to existing data AND confusing to the classifier.

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Resolving Identity Collisions and Triggering VIP Overrides

Case A: Pre-existing Track

Track already has a Global ID. Protected by the Duplicate Guard (`claimed_gids_this_frame`), ensuring two separate bounding boxes in the exact same frame cannot share an ID.



Case B: Face Override

If ArcFace identifies a face, it instantly bypasses body matching.

If the face belongs to GID 29 but the body track thought it was GID 26, an `execute_identity_merge` rewires MongoDB, moves local files, and collapses GID 26 into GID 29.

Case C: Body Classification

Computes weighted centroids.

Requires `Cosine Similarity > 0.78` AND a `Softmax Ratio (Tau) >= 2.0`.

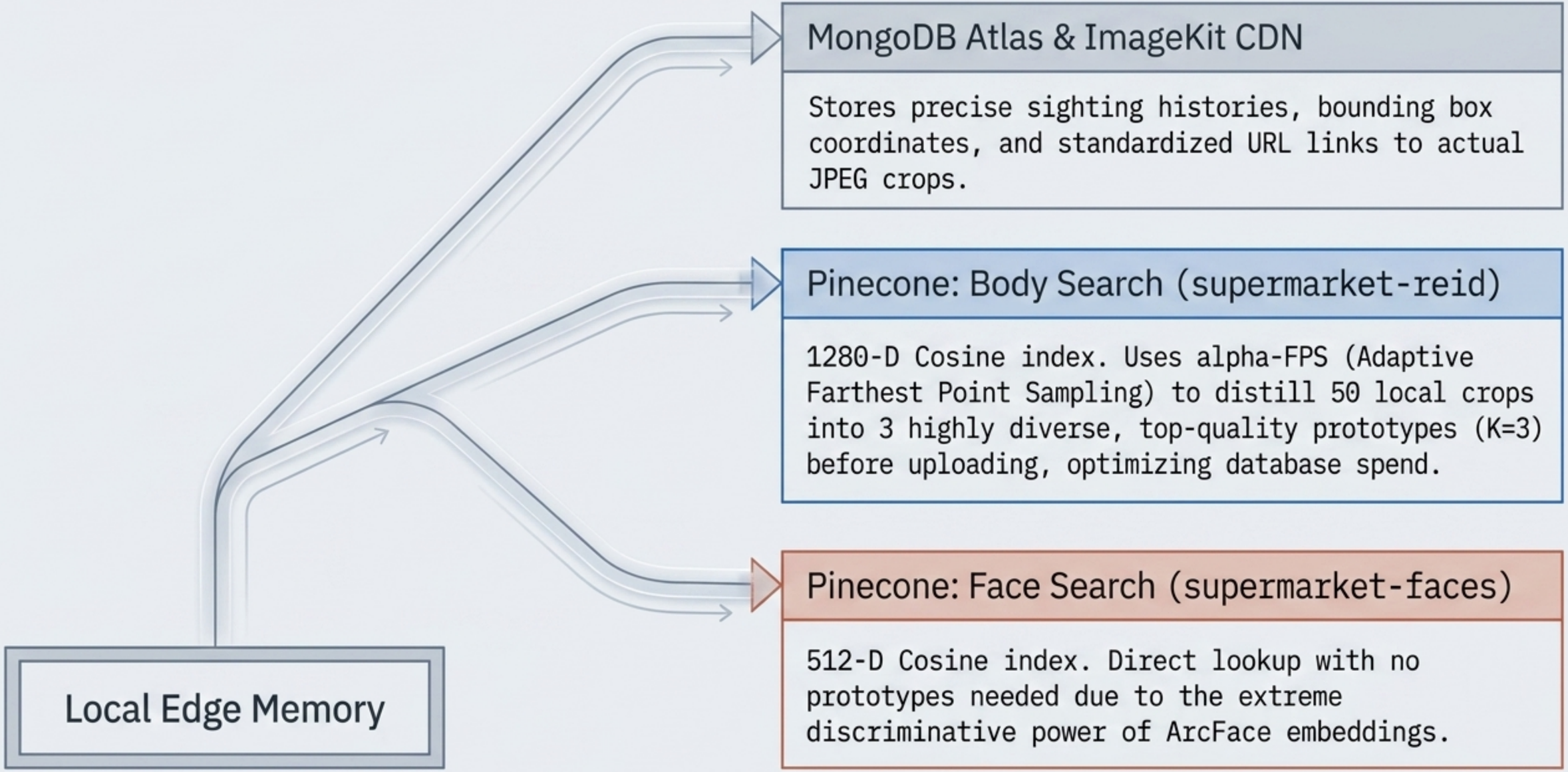
Case D: Unknown Data Manager

Ambiguous items go to the UDM.

Every 30 frames, `DBSCAN` **clustering** (`Epsilon = 0.15`) attempts to mint new IDs or merge clusters based on a minimum size of 5 crops.

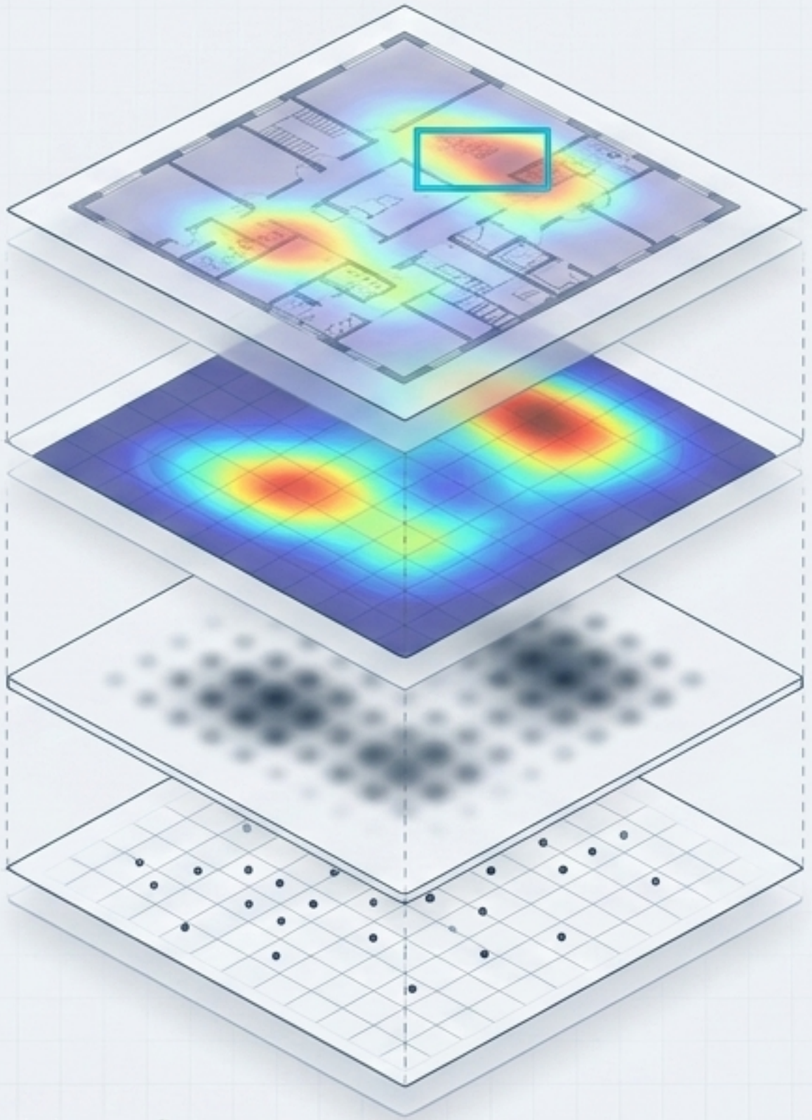
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Scaling Local Memory into Multimodal Cloud Architecture



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Rendering Spatial Sighting Histories into Movement Heatmaps



Layer 4: Final Output Blend

Final output blended onto the reference PNG background (Alpha=0.6) with a Cyan Rectangle marking the last known position.

Layer 3: Upscaled Colorized Overlay

`cv2.COLORMAP_JET` applied to the density estimation.

Layer 2: Downscaled Blurred Canvas

Stamps applied, Gaussian blur Sigma=5.

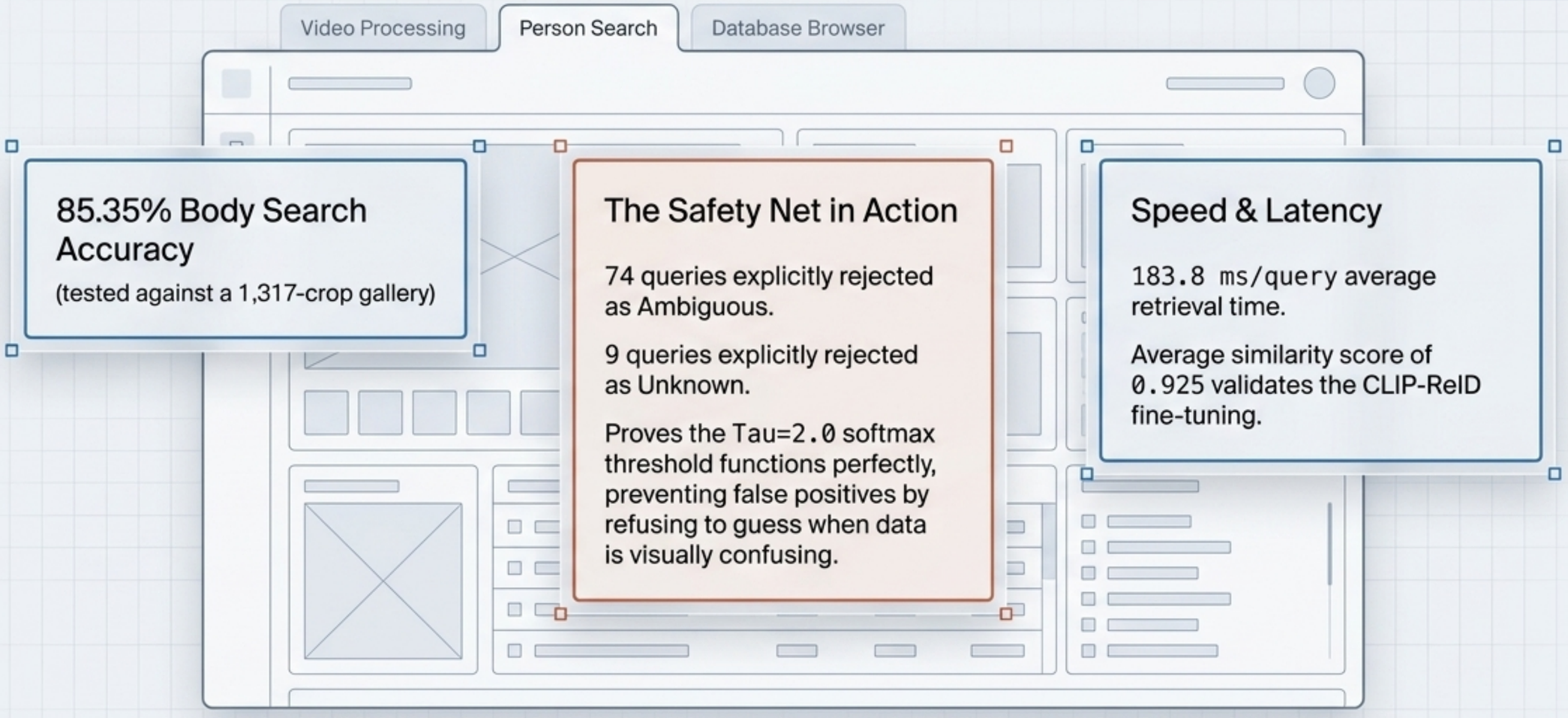
Layer 1: BBox Centers

Raw data points from MongoDB `best_crop_bbox`.

Two-pass approach evaluates the Gaussian density on a 10% downscaled canvas first, saving massive computational overhead before upscaling via linear interpolation.

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Validating the Pipeline with a Safety-First Approach



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Engineering Trade-offs, Edge Cases, and System Constraints

⚠ RAM-First Vulnerability

Sightings are accumulated in memory and uploaded in bulk at video completion. Optimized for Kaggle/Colab IOPs limits, but a mid-processing crash results in **total session data loss**.

⚠ Database Synchronization Disconnects

A local `RESET_DATABASE` wipes MongoDB and local pickles, but explicitly does NOT touch ImageKit or Pinecone, requiring manual vector pruning.

⚠ Heatmap Dependencies

Spatial accuracy fundamentally depends on an exact resolution match between the source video and the `bg_{camera_id}.png`. Missing backgrounds force the system into a **dark-canvas fallback**, losing real-world spatial context.

⚠ Single-Frame Fragmentation

If `BotSort` fragments one person into two temporary tracks, the Duplicate Guard forces one into the Unknown pool--an intentional, fail-safe behavior to protect the main identity.

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